2017 Winter Qual

III. $A_1 + \cdots + A_k = I$

$$A_1 A_i + \cdots + A_r A_r + \cdots + A_k A_r = A_r$$

$$A_r A_1 A_r + \cdots + \cancel{A_r} + \cdots + A_r A_k A_r = \cancel{A_r}$$

$$\underbrace{x^T A_r}_{= y^T} A_1 \underbrace{A_r x}_{= y} + \cdots + x^T A_r A_k A_r x = 0 \quad , x \neq 0$$

$$\Rightarrow \underbrace{y^T A_1 y}_{=0} + \cdots + \underbrace{y^T A_k y}_{=0} = 0$$

$$\Rightarrow A_r A_1 A_r = 0, \quad \cdots, \quad A_r A_k A_r = 0$$

$$\Rightarrow (A_1 A_r)^T (A_1 A_r) = 0, \quad \cdots, \quad (A_k A_r)^T (A_k A_r) = 0$$

$$\Rightarrow A_1 A_r = 0, \quad \cdots, \quad A_k A_r = 0$$

$$\Rightarrow A_r A_j = 0 \quad \forall r \neq j$$

$\blacksquare$

四.

$$S_{xy} = \frac{x^T(I-H_1)y}{n-1}$$

$$\mathbb{E}[x^T(I-H_1)y] = \cancel{tr}((I-H_1)\sigma_{xy}) + \mu_x(I-H_1)\mu_y$$

$$= \sigma_{xy}(n-1) + \underbrace{\mu_x 1(I-H_1)1\mu_y}_{=0}$$

$$= \sigma_{xy}(n-1)$$

$$\mathbb{E}[S_{xy}/n-1] = \sigma_{xy} \quad \therefore \text{HE}$$

$\blacksquare$